
ASO SAP Integration Standard

SAP endpoint inventory, APIM/Function wrapper rules, response contract, idempotency, security, and telemetry.

GitHub path: `sap/ASO_SAP_Integration_Standard.md`

Version 1.1 - public customer-ready split package - refined for readability and GitHub publishing

Safety note: this public reference pack must not contain tenant IDs, credentials, secrets, real customer data, private endpoints, or connection strings.

SAP integration standard

SAP S/4HANA is the financial truth, but ASO never connects directly to SAP from AI agents, Power Automate, or Dataverse. All SAP access passes through Azure API Management and Azure Functions.

No autonomous ERP posting by any AI agent. Power Automate may call SAP write endpoints only after approval, idempotency, telemetry, and testing are proven.

SAP preparation

- Create or confirm read-only communication user.
- Create or confirm write-capable communication user for approved commercial submissions only.
- Create communication system for Azure integration.
- Create communication arrangements for required APIs.
- Record arrangement ID, base URL, auth model, service path, owner, and environment.

Endpoint inventory

Capability	Endpoint	Method	Used by	Rule
Business partner search	/sap/business-partner/search	GET	Foundry / lead agents	Read-only through APIM -> Function -> SAP.
Customer profile	/sap/customer-profile	GET	Foundry / opportunity agents	Normalize response before use.
Product affinity	/sap/products/by-customer	GET	Foundry	Used for recommendations.
Sales orders	/sap/sales-orders/by-customer	GET	Foundry	Used for account context.
Account commercial profile	/sap/account-commercial-profile	GET	Foundry	Read-only commercial context.
Prepare order	/sap/commercial/prepare-order	POST	Power Automate after approval path design	Preparation only; validate payload and idempotency.
Submit order	/sap/commercial/submit-order	POST	Power Automate only after approval	Write operation; idempotency required; no blind retry.

Wrapper endpoint response contract

```
{
  "message_id": "uuid",
  "correlation_id": "uuid",
  "status": "success|failed|partial_success",
  "data": {},
  "error": {
```

```
    "code": "",
    "message": "",
    "retryable": true,
    "origin": "sap-wrapper"
  },
  "trace_id": "string"
}
```

Idempotency rules

- Every write-path request must include an idempotency key.
- The idempotency key must be stored with the Pending Commercial Action.
- Do not blindly retry create/submit operations without idempotency verification.
- Commercial submit requires human approval before APIM call.
- SAP document references must be written back to Dataverse after successful response.

Security and observability

- Function App uses managed identity where possible.
- Secrets live in Key Vault, not Power Automate or GitHub.
- APIM validates JWT and passes correlation headers.
- Application Insights logs request counts, dependency duration, timeout rate, and failures.
- Failed submissions preserve payload, error message, trace ID, and create Integration Operator task.